

SCH4U Unit Test 3

K/U	A	T	C	Total
51	8	11	10	80

Multiple Choice

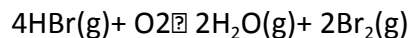
Identify the letter of the choice that best completes the statement or answers the question

- In an exothermic reaction, the temperature of the surroundings increases because the molecules in the surrounding have [3]
 - A lower kinetic energy
 - A lower potential energy
 - A greater kinetic energy
 - A greater potential energy
 - Not changed in either kinetic or potential energy
- The molar heat of vaporization of water is 42kJ/mol. How much energy is released by the condensation of 3.0g of water?[3]
 - 0.88kJ
 - 7.0kJ
 - 130kJ
 - 250kJ
 - 0.07kJ
- Which of the following reactions is exothermic?[3]
 - $\text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) \Delta\text{H} = -243\text{kJ/mol}$
 - $1/2 \text{N}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) + 90.7\text{kJ} \rightarrow \text{NO}(\text{g})$
 - $\text{CO}(\text{g}) + 111\text{kJ} \rightarrow \text{C}(\text{s}) + 1/2 \text{O}_2(\text{g})$
 - $1/2 \text{H}_2(\text{g}) + 1/2 \text{I}_2(\text{g}) \rightarrow \text{HI}(\text{g}) \Delta\text{H} = +26\text{kJ/mol}$
 - $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) - \Delta\text{H} = -243\text{kJ/mol}$
- Consider the following equation for the combustion of hydrogen:
 $\text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) + 243\text{kJ}$
In order to produce 1215kJ of heat, how many grams of H_2 must burn?[3]
 - 12.0g
 - 0.250g
 - 0.100g
 - 8.00g
 - 10.0g

5. What is the average rate of production of $\text{H}_2(\text{g})$ in the reaction of zinc with hydrochloric acid if after 30s the volume of hydrogen is 20mL, and after 1 minute, the volume is 62mL?[3]
 - a. 62mL/s
 - b. 1.4mL/s
 - c. 62mL/min
 - d. either a or b
 - e. either b or c
6. Generally, temperature affects the rate of a reaction in which of the following ways?[3]
 - a. Increasing the temperature reduces the rate of the reaction
 - b. Decreasing the temperature decreases the rate of the reaction
 - c. Increasing the temperature increases the rate of the reaction
 - d. Both a and b are correct
 - e. Both b and c are correct
7. Using the standard heats of formation, what is the enthalpy change per mole of CO_2 for the following reaction?[3]

$$\text{CO}(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$$
 - a. -506kJ
 - b. -284kJ
 - c. +284kJ
 - d. +506kJ
 - e. -617kJ
8. Based on the heat of formation of MgO , how much energy is required to decompose 48.6g of MgO ?[3]
 - a. 603kJ
 - b. 725kJ
 - c. 301kJ
 - d. 60.3kJ
 - e. 1504kJ
9. A catalyst is only effective if it[3]
 - a. Is present in large quantities
 - b. Is added before the reactants come in contact with each other
 - c. Is specific to the particular reaction
 - d. Is heated before it is added to the reaction
 - e. All of the above
10. If for the reaction, $a\text{X} + b\text{Y} \rightarrow \text{products}$, the rate law is determined to be $r = k[\text{x}]^1[\text{y}]^2$, then the order of the reaction is which of the following?[3]
 - a. 1
 - b. 2
 - c. 3

- d. 4
 - e. 5
 - f.
11. The half-life of a substance is[3]
- a. Half the time for the reactant to be completely consumed
 - b. The time for the reaction to be used up
 - c. The time for all of the reactant to be used up
 - d. The time for half the reactant to be used up
 - e. None of the above
12. Which of the following is not part of the collision theory?[3]
- a. A chemical system consists of particles that are in constant motion
 - b. A reaction must involve collisions of particles
 - c. An effective collision has sufficient energy and correct orientation
 - d. An ineffective collision can still cause a reaction
 - e. The rate of the reaction depends on the frequency of effective collisions
13. The activated complex[3]
- a. is an unstable molecule
 - b. has the maximum potential energy possible
 - c. may continue on to produce products
 - d. may revert to reactants
 - e. all of the above
14. When a reaction is broken down into a number of elementary steps, this is known as a [3]
- a. reaction progression
 - b. reaction mechanism
 - c. reaction
 - d. reaction step
 - e. none of the above
15. Reaction intermediates are [3]
- a. steps within a reaction mechanism
 - b. the step that determines the rate of the reaction
 - c. products formed in the overall equation
 - d. products formed within some steps of the reaction mechanism
 - e. none of the above
1. $\text{HBr(g)} + \text{O}_2\text{(g)} \rightarrow \text{HOBr(g)}$ (slow)
 2. $\text{HOBr(g)} + \text{HBr(g)} \rightarrow 2\text{H}_2\text{O(g)}$ (fast)
 3. $2(\text{H}_2\text{O(g)} + \text{HBr(g)}) \rightarrow \text{H}_2\text{O} + \text{Br}_2\text{(g)}$ (fast)
-



16. Consider the above reaction mechanism. The rate-determine step of this reaction is [3]

- a. elementary step 1
- b. elementary step 2
- c. elementary step 3
- d. elementary steps 2 and 3
- e. impossible to tell from this information

17. The theoretical effect of an increase in the initial concentration of a reactant can be explained in terms of collision theory because it affects [3]

- I. the collision geometry involved in the reaction
- II. the total number of collisions that occur
- III. the fraction of collisions that are effective
- IV. the required activation energy for a reaction

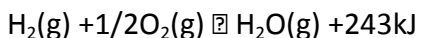
- a. Both I and IV
- b. I, II and III are true
- c. Both II and III
- d. Both III and IV
- e. II only

Short Answer

18. $\Delta H_{\text{combustion}}$ for $\text{C}_9\text{H}_{18} = -5615\text{kJ}$

- a. write the thermochemical equation for the combustion of C_9H_{18} [3T]
- b. Draw Enthalpy diagram for part a [3A]

19. Consider the following equation for the combustion of H₂(hydrogen):



In order to produce 972kJ of heat, how many grams of H₂ must burn?[8T]

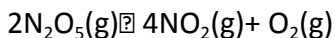
20. A mechanism for a naturally occurring reaction that destroys ozone is:

a. What is the overall reaction?[5C]

b. What are the intermediates?[3C]

c. What is a catalyst?[2C]

21. The decomposition of dinitrogen pentoxide is described by the reaction:



If the rate of appearance of O₂ is equal to 2.40mol/min, what is the rate of disappearance of N₂O₅[5A]

- a. 0.600mol/min
- b. 1.20mol/min
- c. 4.80mol/min
- d. 9.60mol/min